

Nikolaos Goutzoulis

EDUCATION Phone: (646)-957-0924 | Email: ng2985@columbia.edu | LinkedIn: Nikos Goutzoulis

Columbia University

Computer Science MS

• GPA: 3.52/4.00

• Relevant Courses: Unsupervised Learning, Advanced Deep Learning, Advanced Big Data and AI

Aug 2024 - Dec 2025

New York, NY

National Technical University of Athens

Electrical and Computer Engineering - Integrated MEng

• Grade: 8.7/10 (top 5% of 400 students)

• Relevant Courses: Deep Learning, Databases, Advanced Algorithms

Jul 2024

Athens, GR

PROFESSIONAL EXPERIENCE

Ripple

Blockchain Data Scientist Intern (Incoming at RippleX team)

May 2025 - Aug 2025

New York, NY

Columbia University

Course Assistant - Analysis of Algorithms

Jan 2025 - Present

New York, NY

• Guiding 150+ students in mastering data structures, **dynamic and linear programming**, and **nonlinear optimization** techniques, emphasizing their application in problem-solving and system design.

• Demonstrating leadership and **communication skills** by holding Q&A sessions and simplifying complex algorithms.

Columbia University

Graduate Research Assistant – Software Systems Lab

Sep 2024 - Apr 2025

New York, NY

• Collaborated with Meta researchers through **Git** to create a **browser API** in **C++** for Chrome by measuring user engagement without cross-site tracking and enabling secure attribution and analytics.

• Designed privacy-aware **data pipelines** across Chrome, integrating differential privacy metrics to support attribution analytics and behavioral modeling at scale.

EmDoT S.A.

Machine Learning Engineer - Master Thesis Collaboration

Oct 2023 - Jul 2024

Athens, GR

• Engineered deep learning models with **PyTorch** using **CUDA**, and machine learning models such as random forests with **Sklearn**, boosting prediction accuracy by 15% for resource allocation KPIs in an insurance company.

• Optimized model performance by 20% and accuracy by performing **feature engineering**, incorporating parameters such as time of day and custom metrics.

• Engaged with cross-functional teams, directly **interacted with customers** and **deployed ML models** in industrial use.

EmDoT S.A.

Data Science Intern

Jul 2023 - Sep 2023

Athens, GR

• Designed algorithms for smoothing Bin Fill Level data, using **SQL**, **Tableau**, and statistical models like **Prophet** and **SARIMAX** to predict the trend of the data.

• Reduced noise in data by 20% while maintaining 95% of original signal's variance.

PUBLICATIONS AND PROJECTS

SOSP (paper under review): Privacy Budget Management in W3C's Privacy-Preserving Attribution API

Apr 2025

Columbia University

• Initiated browser API integration in Chrome and built a Firefox extension to visualize differential privacy budgets.

NutritionAI: LLM Nutritional Personalized Assistant

Jan 2025 - Mar 2025

Columbia University

• Preprocessed a conversational and nutritional dataset with **Python scripts** and fine tuned **Llama3 8B** on AWS's NVIDIA T4 GPU. The natural language processing project produced personalized nutritional programs for each user.

Evaluating LLM Resilience to Adversarial Data Poisoning

Sep 2024 - Dec 2024

Columbia University

• Designed a method to assess the resilience of **RoBERTa LLM** in sentiment analysis against adversarial data poisoning, which revealed a model accuracy drop of up to 15%.

• Generated text embeddings using BERT and performed **dimensionality reduction** within **Databricks** to facilitate sentiment analysis.

Aegean Shipping Management Scholarship

Jul 2024

SKILLS AND CERTIFICATIONS

Programming Languages: Python, C++, JavaScript, Java, Matlab, **SQL**, Rust, R

ML Frameworks: Tensorflow, CUDA, PyTorch, Keras, Scikit-learn, Transformers, XGBoost

Cloud, Development and Other Tools: GCP, Databricks, Docker, Kubernetes, PowerBI, API design, Visual Studio,

Matplotlib, Git, Tableau, React, Node.js